

INTERVIEW PREPARATION · LENDING & FINTECH

The Loan Lifecycle

A complete, interview-ready field guide — every stage a loan passes through, the financial math behind it, the regulations that govern it, the systems that run it, and the questions you will be asked about all of it.

Covers: Origination · Underwriting · Pricing · Closing & Funding · Servicing · Delinquency & Collections · Loss Mitigation · Charge-off & Recovery · Payoff

Plus: Amortization math · Key ratios · Loan types · U.S. regulation · System-design considerations · 40+ interview Q&A · Glossary

Audience: Candidates for product, engineering, data, risk, and ops roles at banks, credit unions, and lending fintechs.

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1 · The lifecycle at a glance

A **loan lifecycle** is the end-to-end journey of a loan from the moment a borrower applies to the moment the debt is fully resolved — either paid off or written off. Every lending business, whether a 150-year-old bank or a two-year-old fintech, runs some version of this pipeline. Knowing the stages, who owns each one, and where money and risk move is the backbone of any lending interview.



Two things to keep straight from the start:

- **The happy path is short.** Most loans go Application → Underwriting → Funding → Servicing → Payoff. Stages 6–8 (delinquency, loss mitigation, charge-off) only happen when a borrower stops paying.
- **"Origination" has two meanings.** Narrowly, it's the application/intake step. Broadly, "origination" refers to everything *before* funding (application through closing). Clarify which one an interviewer means.

STAGE	OWNER	CORE QUESTION BEING ANSWERED
Origination	Sales / Loan officer	Who is this borrower and what do they want?
Underwriting	Underwriter / Risk model	Can and will they repay?
Pricing	Risk / Capital markets	What rate compensates us for the risk?
Closing & Funding	Ops / Closing	Are documents signed and money disbursed?
Servicing	Servicing / Ops	Are payments coming in correctly?
Collections	Collections	How do we recover a missed payment?
Loss mitigation	Loss mit / Workout	Can we restructure to avoid loss?
Charge-off / Recovery	Recovery / Legal	Accept the loss and recover what we can.

2 · Stage 1 — Origination & application

The borrower expresses intent and provides data. The lender captures the application, verifies identity, and pulls the information needed to make a decision.

What happens

- **Application intake.** Borrower submits personal details, requested amount, term, and purpose. Channels: branch, call center, web, mobile, broker, or point-of-sale (e.g., checkout financing).
- **Identity verification (KYC).** "Know Your Customer" — confirm the person is real and who they claim to be. Tied to **AML** (Anti-Money-Laundering) and **CIP** (Customer Identification Program) requirements, and OFAC sanctions screening.
- **Consent & disclosures.** Borrower authorizes a credit pull and receives initial disclosures (for mortgages, the Loan Estimate within 3 business days).
- **Data collection.** Credit report/score, income documents (pay stubs, W-2s, bank statements, tax returns), employment verification, and for secured loans, collateral details.
- **Pre-qualification vs. pre-approval.** *Pre-qual* is a soft, self-reported estimate (soft credit pull, no commitment). *Pre-approval* is a stronger conditional offer based on a hard pull and verified data.

WATCH OUT

Soft vs. hard credit inquiry. A *soft pull* doesn't affect the borrower's score and is used for pre-qualification and marketing. A *hard pull* can lower the score a few points and requires borrower permission (FCRA "permissible purpose"). Doing a hard pull without consent is a compliance violation.

Concepts to know

TERM	MEANING
KYC / CIP	Verifying borrower identity to prevent fraud and money laundering.
Credit bureaus	Equifax, Experian, TransUnion — they compile credit reports. FICO & VantageScore turn reports into scores.
Tri-merge	A combined report pulling all three bureaus (common in mortgage).
Stips	"Stipulations" — documents the lender requires before approval (proof of income, ID, etc.).

3 · Stage 2 — Underwriting

The risk decision. Underwriting assesses whether the borrower **can** repay (capacity) and **will** repay (willingness), and whether the loan fits the lender's policy and risk appetite.

The Five Cs of Credit

The classic framework — expect to be asked to name them:

C	WHAT IT MEASURES	TYPICAL SIGNAL
Character	Willingness to repay; credit history.	Credit score, payment history, derogatory marks.
Capacity	Ability to repay from income.	Debt-to-income (DTI), cash flow.
Capital	Borrower's own skin in the game.	Down payment, savings, reserves.
Collateral	Asset securing the loan.	Loan-to-value (LTV), appraisal.
Conditions	Loan purpose & the economic environment.	Rate environment, industry, use of funds.

Manual vs. automated underwriting

- **Manual underwriting.** A human underwriter reviews the file, exercises judgment, handles exceptions and complex income. Slower, costlier, used for jumbo/complex loans.
- **Automated underwriting systems (AUS).** Rules + models decide instantly. In mortgage: Fannie Mae's *Desktop Underwriter (DU)*, Freddie Mac's *Loan Product Advisor (LPA)*. Fintech lenders use proprietary ML scorecards.
- **Hybrid.** Auto-approve the clean middle, auto-decline the clear no's, route the gray zone to humans. This "refer" bucket is where most interview system-design discussion lives.

WATCH OUT — FAIR LENDING

Underwriting models must not discriminate against protected classes (ECOA / Reg B). Even if a model never sees race, a feature like ZIP code can act as a **proxy** and create **disparate impact**. Models need fair-lending testing, adverse-action reasons, and explainability. This is a favorite interview probe for data/ML roles.

Outputs of underwriting

- **Approve** — proceed, possibly with conditions ("conditional approval" / "approved with stip").
- **Decline** — must send an **adverse action notice** (ECOA/FCRA) stating the principal reasons.
- **Counter-offer** — approve a smaller amount, shorter term, or higher rate.

4 · Stage 3 — Approval & pricing

An approved loan still needs a **price** — the interest rate and fees. Pricing translates risk into a number.

Risk-based pricing

Higher-risk borrowers pay higher rates. Conceptually the rate is built up from components:

COMPONENT	COVERS
Cost of funds	What it costs the lender to borrow/raise the money.
Operating cost	Origination & servicing overhead.
Expected loss	$PD \times LGD \times EAD$ — the credit risk premium (see §12).
Cost of capital	Return required on the equity held against the loan.
Profit margin	Target spread.

Rate vs. APR

Interest rate is the cost of borrowing the principal, expressed annually. **APR (Annual Percentage Rate)** folds in points and most fees, so it's the true all-in cost and is required by TILA for an apples-to-apples comparison. **APR** $\geq$ **interest rate** almost always. APY (yield) is the savings-side analog that includes compounding.

Fixed vs. variable

- **Fixed rate** — same rate for the life of the loan; predictable payment.

- **Variable / adjustable (ARM)** — rate = *index* (e.g., SOFR, Prime) + *margin*, resetting periodically, often with caps on each adjustment and a lifetime cap.

In mortgage, the rate can be "locked" for a window so it doesn't move between approval and closing. Points are optional upfront fees paid to buy down the rate.

5 · Stage 4 — Closing & funding

The loan becomes legally binding and the money moves.

- **Final disclosures.** For mortgages, the **Closing Disclosure** must reach the borrower at least **3 business days before** closing (TRID rule), so they can compare it against the Loan Estimate.
- **Document execution.** Borrower signs the **promissory note** (promise to repay) and, for secured loans, the **security instrument** (mortgage/deed of trust, or lien on the vehicle title).
- **Perfecting the lien.** Recording the mortgage or filing a **UCC-1** makes the lender's claim on collateral enforceable and senior.
- **Funding / disbursement.** Money is wired to the borrower, dealer, seller, or paid to the title/escrow company. The **funding date** often differs from the signing date.
- **Right of rescission.** For certain refinances/home-equity loans, the borrower gets 3 business days to cancel before funds disburse.
- **Boarding.** The new loan is "boarded" onto the servicing system — account number, amortization schedule, and first-payment date created.

NOTE · ORIGINATION ENDS HERE

Everything up to and including funding is the **origination** side of the house. After boarding, the loan crosses into **servicing**. Many fintechs originate loans and then sell them; servicing may be done in-house or by a third-party *sub-servicer*.

6 · Stage 5 — Servicing

The long, quiet middle. Servicing is everything that keeps a performing loan running until payoff — usually years. This is where most of a lending platform's day-to-day software lives.

Core servicing functions

- **Payment processing.** Accept payments (ACH, card, check, wire), apply them in the correct order, and post to the ledger.
- **Payment application waterfall.** A payment is typically applied *fees* → *interest* → *principal*, or in many consumer loans *interest* → *principal* → *fees*. Order matters and is governed by the note. Extra payments may go

to principal ("principal curtailment").

- **Interest accrual.** Interest accrues daily on the outstanding principal (see §11). Servicers run nightly accrual jobs.
- **Amortization schedule.** The plan of how each payment splits between interest and principal over the term.
- **Escrow (mortgage).** Servicer collects 1/12 of annual property taxes & insurance with each payment and pays those bills when due. Annual escrow analysis trues up shortages/surpluses.
- **Statements & notices.** Periodic statements, year-end interest (Form 1098), rate-change notices for ARMs.
- **Customer servicing.** Payoff quotes, due-date changes, autopay setup, hardship intake.

Statuses a performing loan moves through

Current → In grace period → Late fee assessed → Delinquent (30/60/90 DPD)

DPD = "days past due." A grace period (often 10–15 days) lets a late payment land without a fee; the loan only reports delinquent to bureaus after ~30 days past due.

Servicing transfers & the secondary market

Loans (or just the servicing rights, "MSRs" for mortgages) are frequently sold. When servicing transfers, the borrower gets a **goodbye/hello letter** and a 60-day window where a payment to the old servicer can't be counted late (RESPA). Loans are also **securitized** — bundled into pools (MBS/ABS) and sold to investors, which is why origination and servicing are often separate businesses.

7 · Stage 6 — Delinquency & default

A borrower misses payments. The loan moves from collections-light reminders to formal default handling.

The delinquency ladder

BUCKET	WHAT'S HAPPENING
1–29 DPD	Early/soft collections: reminders, texts, calls. Often curable quickly.
30 DPD	First bureau-reportable delinquency. Late fee already assessed.
60–90 DPD	Escalating contact. Loss-mitigation options offered.
90+ DPD	Serious delinquency. Often the trigger for default classification.
Default	Borrower has breached the note's terms. Acceleration may be invoked.

Key concepts

- **Delinquent vs. default.** *Delinquent* = behind on a payment. *Default* = a formal breach of the contract (definition is in the note; commonly 90+ days, but can include other covenant breaches).
- **Acceleration.** After default, the lender can declare the *entire* remaining balance due immediately — not just the missed payment.
- **Cure.** Bringing the loan current by paying the past-due amount (plus fees) returns it to performing status.
- **Roll rates.** The probability a loan in one bucket "rolls" to the next (30 → 60 → 90). Core to loss forecasting and collections staffing.

WATCH OUT — FDCPA

Collections activity is tightly regulated by the **FDCPA** (and state law): no calls before 8am/after 9pm, no harassment, no false threats, honor cease-contact requests. This applies most strictly to third-party debt collectors but shapes first-party practice too.

8 · Stage 7 — Loss mitigation (workout)

Foreclosing or charging off is expensive. Loss mitigation tries to keep the borrower paying *something* and minimize the lender's loss. Expect to be asked the difference between these tools:

OPTION	WHAT IT DOES
Forbearance	Temporarily pause or reduce payments during hardship; arrears repaid later. Terms unchanged.
Deferment	Move missed payments to the end of the loan. Common in auto/student loans.
Repayment plan	Spread the past-due amount over several months on top of regular payments.
Loan modification	Permanently change terms — lower rate, longer term, or principal forbearance/forgiveness — to make payments affordable.
Refinance	Replace the old loan with a new one on better terms (borrower must still qualify).
Short sale / DIL	For secured loans: sell the collateral for less than owed, or hand it back ("deed in lieu") to avoid foreclosure.

TDR — KNOW THIS TERM

When a lender grants a concession to a troubled borrower, it may be classified as a **Troubled Debt Restructuring**, which carries specific accounting and reporting treatment. Under newer accounting standards (CECL), TDR-specific designation was eliminated in favor of broader expected-loss disclosure — a nice detail to mention for risk/accounting-adjacent roles.

9 · Stage 8 — Charge-off & recovery

When recovery looks unlikely, the lender accepts the loss on its books — but the debt doesn't vanish.

- **Charge-off.** An *accounting* action: the lender declares the debt unlikely to be collected and removes it from assets (moves it against the loan-loss reserve). For most consumer loans this happens at **120–180 DPD** (regulatory guidance). *The borrower still legally owes the money.*
- **Recovery.** After charge-off, the lender still tries to collect — in-house, via a contingency collection agency, or by **selling the debt** to a debt buyer for cents on the dollar. Anything recovered is "recovery income."
- **Repossession / foreclosure.** For secured loans, the lender seizes and sells the collateral. Proceeds reduce the balance; any remaining gap is a **deficiency balance** the borrower may still owe.
- **Provision & reserves.** Lenders set aside an **allowance for loan losses** in advance (under **CECL**, based on *expected lifetime losses*). Charge-offs draw down this reserve; the periodic top-up is the "provision expense" on the income statement.

WATCH OUT — COMMON CONFUSION

Charge-off ≠ forgiveness. A charged-off debt can still be collected, sold, and reported on credit for up to 7 years. It only becomes uncollectible-by-law when the *statute of limitations* passes or the debt is formally settled/discharged (e.g., bankruptcy).

10 · Stage 9 — Payoff & closure

The happy ending, and the most common one. The loan is fully satisfied.

- **Payoff quote.** Outstanding principal + accrued interest to the payoff date + any fees. It changes daily because interest accrues — quotes are "good through" a specific date.
- **Routes to payoff:** scheduled final payment, early full prepayment, refinance (new loan pays off old), or sale of the collateral.
- **Prepayment penalty.** Some loans charge a fee for paying off early (lender loses expected interest). Many consumer loans prohibit them; disclosure is required where allowed.
- **Lien release.** For secured loans, the lender files a release/satisfaction (mortgage satisfaction, lien release on title) so the borrower owns the asset free and clear.
- **Account closure & reporting.** Loan marked "paid/closed" and reported to bureaus as paid in full — a positive credit event.

11 · The money math

You will likely be asked to reason about at least one of these. Know the intuition even if you don't memorize the formula.

Amortization

An amortizing loan has a fixed periodic payment. Early payments are mostly interest; later payments are mostly principal. The fixed payment is:

$$M = P \cdot [i(1+i)^n] / [(1+i)^n - 1]$$

M = payment · P = principal · i = periodic rate (annual ÷ periods) · n = number of payments

Each period: interest = balance × i , principal = M - interest, new balance = balance - principal. Repeat until zero.

MONTH	PAYMENT	INTEREST	PRINCIPAL	BALANCE
—	\$10,000 loan · 12% APR (1%/mo) · 12 months			\$10,000.00
1	\$888.49	\$100.00	\$788.49	\$9,211.51
2	\$888.49	\$92.12	\$796.37	\$8,415.14
...	...	...	...	...
12	\$888.49	\$8.80	\$879.69	\$0.00

Notice interest shrinks and principal grows each month — that's amortization in action.

Interest accrual conventions

- **Simple daily interest:** daily accrual = (balance × annual rate) / 365. Most auto/personal loans. Paying early saves interest; paying late costs more.
- **Day-count conventions:** 30/360, actual/365, actual/360 — they change how a day's interest is computed. Easy to get wrong in code; a classic source of "off by a few cents" bugs.
- **Compounding:** interest charged on accrued interest. Credit cards compound daily; most installment loans don't compound if paid on schedule.

Interest rate vs. APR vs. APY

METRIC	INCLUDES	USED FOR
Nominal rate	Just interest on principal	Computing the payment
APR	Interest + most fees/points	Comparing loan cost (TILA)
APY / EAR	Interest + compounding effect	True yield on savings/investments

12 · Key ratios & risk metrics

METRIC	FORMULA / MEANING	WHY IT MATTERS
DTI Debt-to-income	Monthly debt ÷ gross monthly income	Capacity to take on payments. Mortgages often cap ~43–50%.
LTV Loan-to-value	Loan amount ÷ collateral value	Cushion if borrower defaults. >80% on a mortgage usually triggers PMI.
FICO / score	300–850 credit score	Summary of credit risk (character).
PD	Probability of Default	Likelihood the loan defaults.
LGD	Loss Given Default (1 – recovery rate)	Share of exposure lost if it defaults.
EAD	Exposure at Default	Balance at risk when default hits.
Expected Loss	$EL = PD \times LGD \times EAD$	The cornerstone of credit pricing & reserves.
NIM Net interest margin	$(\text{Interest income} - \text{interest expense}) \div \text{assets}$	Core lending profitability.
NCO Net charge-off rate	$(\text{Charge-offs} - \text{recoveries}) \div \text{avg loans}$	Realized credit losses; portfolio health.
Delinquency rate	Balance 30+ DPD ÷ total balance	Leading indicator of future losses.
Vintage / cohort	Loans grouped by origination period	Compares how each "vintage" performs over its life.

13 · Loan types (and how the lifecycle bends)

Mortgage

Secured by real estate. Long (15–30 yr), heavily regulated (TRID, RESPA), escrow for taxes/insurance, foreclosure on default.

Auto loan

Secured by the vehicle. Shorter term, fast (often instant) underwriting, repossession on default, deficiency balances common.

Personal / installment

Usually unsecured. Fixed payments; fintech sweet spot. Charge-off (no collateral) rather than repossession.

Credit card (revolving)

No fixed term; a credit *line*. Minimum payments, daily compounding, utilization matters. "Lifecycle" loops continuously.

Student loan

Federal vs. private differ hugely. Deferment/forbearance, income-driven repayment, special bankruptcy treatment.

Small business / commercial

Underwriting on business financials & cash flow; often personal guarantees, covenants, UCC liens.

HELOC

Secured vs. unsecured is the single biggest fork: secured loans add collateral valuation, lien perfection, and repossession/foreclosure; unsecured loans lean harder on credit and income, and go straight to charge-off.

Revolving line secured by home equity. Draw period then repayment period.

BNPL

"Buy Now Pay Later." Tiny, short, point-of-sale installments. Light underwriting; growing regulatory attention.

14 · U.S. regulatory landscape

You don't need to be a lawyer, but naming the right acronym at the right stage signals real domain knowledge.

LAW / REG	WHAT IT GOVERNS	STAGE
TILA (Reg Z)	Truth in Lending — APR & cost disclosures, right of rescission.	Pricing / Closing
RESPA (Reg X)	Real Estate Settlement — closing costs, escrow, servicing transfers.	Closing / Servicing
TRID	TILA-RESPA Integrated Disclosure — Loan Estimate & Closing Disclosure.	Origination / Closing
ECOA (Reg B)	Equal Credit Opportunity — no discrimination; adverse-action notices.	Underwriting
FCRA	Fair Credit Reporting — credit pulls, accuracy, permissible purpose.	Origination / Servicing
FDCPA	Fair Debt Collection Practices — how debts may be collected.	Collections
HMDA	Home Mortgage Disclosure — report lending data for fair-lending oversight.	Reporting
BSA / AML	Bank Secrecy / Anti-Money-Laundering — KYC, CIP, SARs.	Origination
UDAAP	Unfair, Deceptive, or Abusive Acts & Practices — catch-all consumer protection.	All stages
SCRA	Servicemembers Civil Relief — rate caps & protections for active military.	Servicing / Collections

*Federal enforcers: the **CFPB** (consumer protection), OCC, FDIC, Federal Reserve, and state regulators.*

15 · System-design considerations

For engineering/data roles, the lifecycle maps onto a set of systems. Be ready to whiteboard one.

The system map

SYSTEM	OWNS
LOS — Loan Origination System	Application, document collection, underwriting workflow, decisioning up to funding.
Decision engine	Rules + ML scorecards; returns approve/decline/refer + price + reasons.
LMS — Loan Management/Servicing System	Ledger, accrual, payment processing, statements, payoff.
Collections platform	Delinquency workflows, dialer/comms, promise-to-pay.
Data warehouse / feature store	Reporting, vintage analysis, model features, regulatory reporting.

Hard problems interviewers love

- **The ledger must be correct and auditable.** Use double-entry bookkeeping, an append-only event log, and idempotent posting. Never mutate balances in place — derive them from immutable transactions. Money bugs are the worst bugs.
- **Idempotency.** A payment webhook or retry must not double-charge or double-post. Idempotency keys everywhere money moves.
- **Money representation.** Never use floats for currency — use integer minor units (cents) or fixed-precision decimals. Rounding rules must be deterministic and documented.
- **Daily batch vs. real-time.** Interest accrual, delinquency aging, and statement cuts are typically nightly batch jobs; payments and decisions are real-time. Reconciling the two (and handling missed/late batches) is a real design question.
- **State machine.** Model loan status as an explicit state machine (current → delinquent → default → charged-off / paid) with allowed transitions and an audit trail.
- **Time & correctness.** Business days, time zones, cutoff times, backdated payments, and day-count conventions all bite. Make "effective date" a first-class concept.
- **Auditability & compliance.** Every decision needs stored inputs, model version, and adverse-action reasons. Regulators can ask "why was this person declined?" years later.
- **Model governance (ML roles).** Versioning, monitoring for drift, fair-lending testing, explainability (reason codes), and a human-review path for the "refer" bucket.

16 · Interview Q&A bank

Work through these out loud. The strong answers are compressed — expand with examples from your own experience.

Fundamentals

Q

Walk me through the life of a loan from application to payoff.

A

Origination (apply + verify) → underwriting (can/will they repay, the 5 Cs) → pricing (risk-based rate, APR) → closing & funding (sign the note, perfect the lien, disburse) → servicing (process payments, accrue interest, escrow) → if they miss payments: delinquency → loss mitigation → charge-off/recovery; otherwise → payoff and lien release. The happy path skips the last three.

Q

What's the difference between delinquency, default, and charge-off?

A

Delinquent = behind on a payment (measured in DPD). *Default* = a formal breach of the note's terms, often 90+ DPD, which can trigger acceleration. *Charge-off* = an accounting decision (typically 120–180 DPD) to remove the asset from the books — but the borrower still legally owes it and recovery continues.

Q

Interest rate vs. APR — which is higher and why?

A

APR is higher (or equal). The rate is just interest on principal; APR adds points and most fees, giving the true annualized cost. TILA requires APR so borrowers can compare loans apples-to-apples.

Credit & risk

Q

Name the Five Cs of credit.

A

Character (willingness/credit history), Capacity (ability to repay/DTI), Capital (skin in the game), Collateral (asset securing the loan/LTV), Conditions (purpose & economic environment).

Q

How would you price a loan?

A

Build the rate up: cost of funds + operating cost + expected loss ($PD \times LGD \times EAD$) + cost of capital + margin. Higher-risk borrowers get higher rates (risk-based pricing). The expected-loss term is what links underwriting risk to the price.

Q

What is Expected Loss and how is it computed?

A

$EL = PD \times LGD \times EAD$. Probability of default $\times$ loss given default $(1 - \text{recovery}) \times$ exposure at default. It drives both loan pricing and the loss reserves a lender holds (CECL).

Q

A borrower has a high credit score but a 55% DTI. Approve?

A

Score covers *character*, but DTI is *capacity* — and 55% means more than half of gross income is already committed to debt, above typical limits. I'd lean decline or counter (smaller amount/longer term), and look at compensating factors: reserves, stable high income, low LTV. Good answer shows you weigh multiple Cs, not just the score.

Servicing & collections

Q

How is a payment applied to a loan?

A

Through a waterfall defined by the note — commonly fees → accrued interest → principal (consumer loans often do interest → principal → fees). Extra amounts can curtail principal. Order matters because it changes how fast principal falls and how much interest accrues next period.

Q

What loss-mitigation options would you offer a borrower in hardship?

A

Match the tool to the situation: short-term hardship → forbearance or a repayment plan; payments permanently unaffordable → loan modification (lower rate/longer term); for secured loans facing loss → short sale or deed in lieu. The goal is to minimize the lender's loss while keeping the borrower paying something.

Q

Why might a lender restructure rather than foreclose?

A

Foreclosure/repossession is slow and expensive, recovers less than the balance (high LGD), and hurts the borrower. A modification that keeps a performing loan often yields a better economic outcome — lower loss given default — even at a reduced rate.

System design / engineering

Q

How would you design the ledger for a loan servicing system?

A

Append-only, double-entry, event-sourced. Balances are derived from an immutable transaction log, never mutated in place. Every money movement carries an idempotency key. Currency stored as integer minor units. Nightly accrual posts interest transactions; payments post in real time. Full audit trail for every posting. This gives correctness, replayability, and auditability.

Q

How do you prevent a payment webhook from being processed twice?

A

Idempotency: each payment event carries a unique key; the ledger rejects a second posting with the same key. Combine with a uniqueness constraint at the database level and transactional posting so retries are safe.

Q

Where would you use real-time vs. batch processing?

A

Real-time: underwriting decisions, payment authorization/posting, payoff quotes. Batch (nightly): interest accrual, delinquency aging/status transitions, statement generation, escrow analysis, regulatory reporting. The tricky part is reconciling them and handling a missed or replayed batch deterministically using effective dates.

Compliance & judgment

Q

A model declines an applicant. What must happen next?

A

Send an **adverse action notice** (ECOA/FCRA) with the principal reasons for denial, and the credit score and bureau info used. So the model must produce stored, explainable reason codes — you can't decline someone with an unexplained black box.

Q

Your model doesn't use race, but approval rates differ sharply by race. Problem?

A

Yes — that's potential **disparate impact**. A feature may be a proxy for a protected class (e.g., ZIP code). You'd run fair-lending testing, examine which features drive the gap, seek less-discriminatory alternatives, and document. "We didn't use the variable" is not a defense.

Q

Why are origination and servicing often separate businesses?

A

Loans get sold and securitized. An originator can fund a loan and sell it (or just the servicing rights) into the secondary market, freeing capital to originate more. The buyer/investor owns the cash flows; a servicer is paid a fee to collect. Different economics, skills, and systems.

17 · Glossary

Acceleration

Demanding the full remaining balance after default.

Adverse action

Notice of denial with reasons, required by ECOA/FCRA.

Amortization

Paying down a loan via fixed payments split between interest and principal.

APR

All-in annual cost including fees; required by TILA.

AUS

Automated Underwriting System (e.g., DU, LPA).

Charge-off

Accounting write-down of an uncollectible loan; debt still owed.

Collateral

Asset pledged to secure a loan.

Curtailment

Extra payment applied directly to principal.

DPD

Days past due.

DTI

Debt-to-income ratio (capacity).

EAD

Exposure at default.

Escrow

Servicer-held account for taxes & insurance.

Forbearance

Temporary pause/reduction of payments.

LGD

Loss given default ($1 - \text{recovery rate}$).

Lien

Legal claim on collateral; "perfected" when recorded/filed.

LTV

Loan-to-value ratio.

Modification

Permanent change to loan terms to aid repayment.

MSR

Mortgage servicing rights (tradeable).

NCO

Net charge-off rate.

Origination

Everything from application through funding.

PD

Probability of default.

Promissory note

The borrower's legal promise to repay.

Recovery

Money collected after charge-off.

Rescission

Borrower's right to cancel certain loans within 3 days.

Roll rate

Probability of moving from one delinquency bucket to the next.

Securitization

Bundling loans into securities sold to investors.

Servicing

Managing a loan after funding until payoff.

TDR

Troubled Debt Restructuring.

TRID

Integrated TILA/RESPA mortgage disclosures.

Underwriting

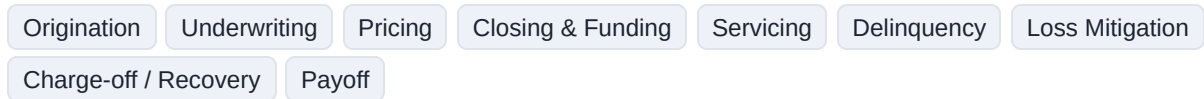
Assessing whether a borrower can and will repay.

Vintage

Loans grouped by origination period for performance analysis.

18 · One-page cheat sheet

The 9 stages



5 Cs of credit

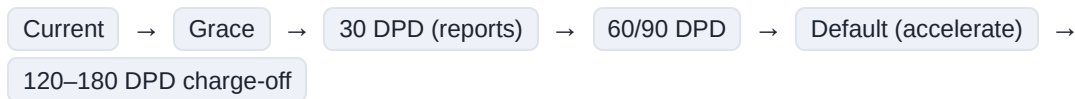


Must-know equations

$EL = PD \times LGD \times EAD$ · $DTI = \text{debt} \div \text{income}$ · $LTV = \text{loan} \div \text{value}$

$M = P \cdot i(1+i)^n / ((1+i)^n - 1)$ · $APR \geq \text{rate}$ · $NCO = (\text{chargeoffs} - \text{recoveries}) \div \text{avg loans}$

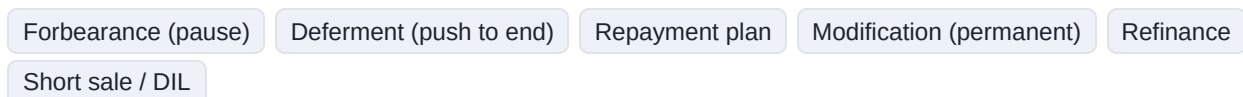
Delinquency ladder



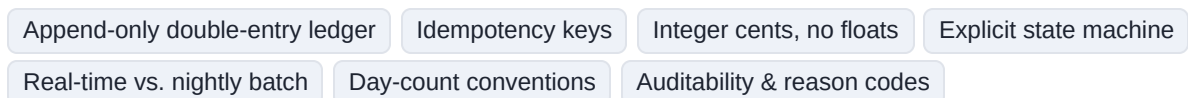
Regulation → stage

KYC/AML/BSA	Origination
ECOA / FCRA	Underwriting (adverse action)
TILA / TRID / RESPA	Pricing & Closing
RESPA / SCRA	Servicing
FDCPA	Collections

Loss-mit toolkit



Engineering watch-list



Prepared as standalone interview-prep material. Educational summary of common U.S. consumer-lending practice; not legal, financial, or compliance advice — specifics vary by product, lender, and jurisdiction.